

Solutions to Questions 1-3

Question 1:

Maximize

$$Z = 2x_1 + x_2$$

subject to

$$6x_1 + 4x_2 \leq 6$$

$$6x_1 + x_2 \leq 3$$

and,

$$x_1 \geq 0, x_2 \geq 0.$$

Introducing slack variables:

Maximize

$$Z = 2x_1 + x_2$$

subject to

$$6x_1 + 4x_2 + x_3 = 6$$

$$6x_1 + x_2 + x_4 = 3$$

and,

$$x_j \geq 0, \text{ for } j = 1, 2, 3, 4.$$

Therefore,

$$\mathbf{c} = [2 \quad 1] \quad [\mathbf{A}, \mathbf{I}] = \begin{bmatrix} 6 & 4 & 1 & 0 \\ 6 & 1 & 0 & 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

Initialization:

$$\mathbf{x}_B = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{B}^{-1}$$

$$\mathbf{c}_B = [0 \quad 0] \quad \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

Calculate the coefficients of nonbasic variables (x_1, x_2) :

$$\mathbf{c}_B \mathbf{B}^{-1} \mathbf{A} - \mathbf{c} = [0 \ 0] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ 6 & 1 \end{bmatrix} - [2 \ 1] = [0 \ 0] - [2 \ 1] = [-2 \ -1]$$

Coefficients of x_1 and x_2 are -2 and -1 respectively.

Since, they have negative coefficients the solution is not optimal.

Iteration 1:

Entering variable: x_1 (variable with most negative coefficient)

Compute the other coefficients of x_1 .

$$\mathbf{B}^{-1} \mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & \dots \\ 6 & \dots \end{bmatrix} = \begin{bmatrix} 6 & \dots \\ 6 & \dots \end{bmatrix} \quad \mathbf{x}_B = \begin{bmatrix} 6 \\ 3 \end{bmatrix} \quad \frac{6}{6} > \frac{3}{6}$$

Therefore, leaving variable: x_4

$$\mathbf{x}_B = \begin{bmatrix} x_3 \\ x_1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 6 \\ 0 & 6 \end{bmatrix} \quad \text{old } \mathbf{B}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 6 \\ 6 \end{bmatrix} \quad \text{new } \mathbf{B}^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & \frac{1}{6} \end{bmatrix}$$

$$\mathbf{c}_B = [0 \ 2] \quad \mathbf{x}_B = \mathbf{B}^{-1} \mathbf{b} = \begin{bmatrix} 1 & -1 \\ 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ \frac{1}{2} \end{bmatrix}$$

Calculate the coefficients of nonbasic variables (x_2, x_4) :

$$\mathbf{c}_B \mathbf{B}^{-1} \mathbf{A} - \mathbf{c} = [0 \ 2] \begin{bmatrix} 1 & -1 \\ 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} \dots & 4 \\ \dots & 1 \end{bmatrix} - [\dots \ 1] = \left[\dots \ \frac{1}{3} \right] - [\dots \ 1] = \left[\dots \ -\frac{2}{3} \right]$$

$$\mathbf{c}_B \mathbf{B}^{-1} = [0 \ 2] \begin{bmatrix} \dots & -1 \\ \dots & \frac{1}{6} \end{bmatrix} = \left[\dots \ \frac{1}{3} \right]$$

Coefficients of x_2 and x_4 are $-\frac{2}{3}$ and $\frac{1}{3}$ respectively.

Since, x_2 has a negative coefficient the solution is not optimal.

Iteration 2:

Entering variable: x_2 (variable with most negative coefficient)

Compute the other coefficients of x_2 .

$$\mathbf{B}^{-1}\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 0 & \frac{1}{6} \end{bmatrix} \begin{bmatrix} \dots & 4 \\ \dots & 1 \end{bmatrix} = \begin{bmatrix} \dots & 3 \\ \dots & \frac{1}{6} \end{bmatrix} \quad \mathbf{x}_B = \begin{bmatrix} 3 \\ \frac{1}{2} \end{bmatrix} \quad \frac{3}{3} < \frac{(1/2)}{(1/6)}$$

Therefore, leaving variable: x_3

$$\mathbf{x}_B = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 4 & 6 \\ 1 & 6 \end{bmatrix} \quad \text{old } \mathbf{B}^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & \frac{1}{6} \end{bmatrix} \quad \begin{bmatrix} 3 \\ \frac{1}{6} \end{bmatrix} \quad \text{new } \mathbf{B}^{-1} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{18} & \frac{2}{9} \end{bmatrix}$$

$$\mathbf{c}_B = [1 \quad 2] \quad \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{18} & \frac{2}{9} \end{bmatrix} \begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{3} \end{bmatrix}$$

Calculate the coefficients of nonbasic variables (x_4, x_3):

$$\mathbf{c}_B\mathbf{B}^{-1} = [1 \quad 2] \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{18} & \frac{2}{9} \end{bmatrix} = \begin{bmatrix} \frac{2}{9} & \frac{1}{9} \end{bmatrix}$$

Coefficients of x_3 and x_4 are $\frac{2}{9}$ and $\frac{1}{9}$ respectively.

Since, both coefficients are non-negative, the current solution $(x_1, x_2, x_3, x_4) = \left(\frac{1}{3}, 1, 0, 0\right)$ is optimal.

$$Z_{opt} = \mathbf{c}_B\mathbf{x}_B = [1 \quad 2] \begin{bmatrix} 1 \\ \frac{1}{3} \end{bmatrix} = \frac{5}{3}.$$

Question 2:

Standard form:

Maximize

$$Z = 2x_1 + x_2$$

subject to

$$x_1 \leq 5$$

$$x_2 \leq 7$$

$$x_1 + x_2 \leq 9$$

and,

$$x_1 \geq 0, x_2 \geq 0.$$

Introducing slack variables:

Maximize

$$Z$$

subject to

$$Z - 2x_1 - x_2 = 0$$

$$x_1 + x_3 = 5$$

$$x_2 + x_4 = 7$$

$$x_1 + x_2 + x_5 = 9$$

and,

$$x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5.$$

Simplex tableaux:

Iteration	Basic Variable	Z	x_1	x_2	x_3	x_4	x_5	RHS
0	Z	1	-2	-1	0	0	0	0
	x_3	0	1	0	1	0	0	5
	x_4	0	0	1	0	1	0	7
	x_5	0	1	1	0	0	1	9
1	Z	1	0	-1	2	0	0	10
	x_1	0	1	0	1	0	0	5
	x_4	0	0	1	0	1	0	7

	x_5	0	0	1	-1	0	1	4
2	Z	1	0	0	1	0	1	14
	x_1	0	1	0	1	0	0	5
	x_4	0	0	0	1	1	-1	3
	x_2	0	0	1	-1	0	1	4

Since, no coefficient in the first row is negative, the current solution

$(x_1, x_2, x_3, x_4, x_5) = (5, 4, 0, 3, 0)$ is optimal.

$$Z_{opt} = 14.$$

Question 3:

Maximize

$$Z = x_1 + 2x_2 + 3x_3 + 4x_4$$

subject to

$$x_1 + 2x_2 + 2x_3 + 3x_4 \leq 20$$

$$2x_1 + x_2 + 3x_3 + 2x_4 \leq 20$$

and,

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0.$$

Introducing slack variables:

Maximize

$$Z = x_1 + 2x_2 + 3x_3 + 4x_4$$

subject to

$$x_1 + 2x_2 + 2x_3 + 3x_4 + x_5 = 20$$

$$2x_1 + x_2 + 3x_3 + 2x_4 + x_6 = 20$$

and,

$$x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5, 6.$$

Therefore,

$$\mathbf{c} = [1 \ 2 \ 3 \ 4] \quad [\mathbf{A}, \mathbf{I}] = \begin{bmatrix} 1 & 2 & 2 & 3 & 1 & 0 \\ 2 & 1 & 3 & 2 & 0 & 1 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 20 \\ 20 \end{bmatrix}$$

Initialization:

$$\mathbf{x}_B = \begin{bmatrix} x_5 \\ x_6 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{B}^{-1}$$

$$\mathbf{c}_B = [0 \ 0] \quad \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 20 \end{bmatrix} = \begin{bmatrix} 20 \\ 20 \end{bmatrix}$$

Calculate the coefficients of nonbasic variables (x_1, x_2, x_3, x_4) :

$$\mathbf{c}_B \mathbf{B}^{-1} \mathbf{A} - \mathbf{c} = [0 \ 0] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 \end{bmatrix} - [1 \ 2 \ 3 \ 4] = [0 \ 0 \ 0 \ 0] - [1 \ 2 \ 3 \ 4] = [-1 \ -2 \ -3 \ -4]$$

Coefficients of (x_1, x_2, x_3, x_4) are $(-1, -2, -3, -4)$ respectively.

Since, they have negative coefficients the solution is not optimal.

Iteration 1:

Entering variable: x_4 (variable with most negative coefficient)

Compute the other coefficients of x_4 .

$$\mathbf{B}^{-1} \mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \dots & 3 \\ \dots & 2 \end{bmatrix} = \begin{bmatrix} \dots & 3 \\ \dots & 2 \end{bmatrix} \quad \mathbf{x}_B = \begin{bmatrix} 20 \\ 20 \end{bmatrix} \quad \frac{20}{3} < \frac{20}{2}$$

Therefore, leaving variable: x_5

$$\mathbf{x}_B = \begin{bmatrix} x_4 \\ x_6 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \quad \text{old } \mathbf{B}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \text{new } \mathbf{B}^{-1} = \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{2}{3} & 1 \end{bmatrix}$$

$$\mathbf{c}_B = [4 \ 0] \quad \mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b} = \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 20 \end{bmatrix} = \begin{bmatrix} \frac{20}{3} \\ \frac{20}{3} \end{bmatrix}$$

Calculate the coefficients of nonbasic variables (x_1, x_2, x_3, x_5) :

$$\begin{aligned}\mathbf{c}_B \mathbf{B}^{-1} \mathbf{A} - \mathbf{c} &= [4 \ 0] \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 & \dots \\ 2 & 1 & 3 & \dots \end{bmatrix} - [1 \ 2 \ 3 \ \dots] \\ &= \begin{bmatrix} \frac{4}{3} & \frac{8}{3} & \frac{8}{3} & \dots \end{bmatrix} - [1 \ 2 \ 3 \ \dots] = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} & \dots \end{bmatrix}\end{aligned}$$

$$\mathbf{c}_B \mathbf{B}^{-1} = [4 \ 0] \begin{bmatrix} \frac{1}{3} & \dots \\ -\frac{2}{3} & \dots \end{bmatrix} = \begin{bmatrix} \frac{4}{3} & \dots \end{bmatrix}$$

Coefficients of (x_1, x_2, x_3, x_5) are $\left(\frac{1}{3}, \frac{2}{3}, -\frac{1}{3}, \frac{4}{3}\right)$ respectively.

Since, x_3 has a negative coefficient the solution is not optimal.

Iteration 2:

Entering variable: x_3 (variable with most negative coefficient)

Compute the other coefficients of x_3 .

$$\mathbf{B}^{-1} \mathbf{A} = \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} \dots & 2 & \dots \\ \dots & 3 & \dots \end{bmatrix} = \begin{bmatrix} \dots & \frac{2}{3} & \dots \\ \dots & \frac{5}{3} & \dots \end{bmatrix} \quad \mathbf{x}_B = \begin{bmatrix} \frac{20}{3} \\ \frac{20}{3} \end{bmatrix} \quad \frac{(20/3)}{(2/3)} > \frac{(20/3)}{(5/3)}$$

Therefore, leaving variable: x_6

$$\mathbf{x}_B = \begin{bmatrix} x_4 \\ x_3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \quad \text{old } \mathbf{B}^{-1} = \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{2}{3} & 1 \end{bmatrix} \quad \begin{bmatrix} \frac{2}{3} \\ \frac{5}{3} \end{bmatrix} \quad \text{new } \mathbf{B}^{-1} = \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix}$$

$$\mathbf{c}_B = [4 \ 3] \quad \mathbf{x}_B = \mathbf{B}^{-1} \mathbf{b} = \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 20 \\ 20 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

Calculate the coefficients of nonbasic variables (x_1, x_2, x_6, x_5) :

$$\mathbf{c}_B \mathbf{B}^{-1} \mathbf{A} - \mathbf{c} = [4 \quad 3] \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} \\ \frac{2}{5} & \frac{3}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix} \begin{bmatrix} 1 & 2 & \dots \\ 2 & 1 & \dots \end{bmatrix} - [1 \quad 2 \quad \dots]$$

$$= \begin{bmatrix} \frac{8}{5} & \frac{13}{5} & \dots \end{bmatrix} - [1 \quad 2 \quad \dots] = \begin{bmatrix} \frac{3}{5} & \frac{3}{5} & \dots \end{bmatrix}$$

$$\mathbf{c}_B \mathbf{B}^{-1} = [4 \quad 3] \begin{bmatrix} \frac{3}{5} & -\frac{2}{5} \\ \frac{2}{5} & \frac{3}{5} \\ -\frac{2}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} \frac{6}{5} & \frac{1}{5} \end{bmatrix}$$

Coefficients of (x_1, x_2, x_5, x_6) are $\left(\frac{3}{5}, \frac{3}{5}, \frac{6}{5}, \frac{1}{5}\right)$ respectively.

Since, all coefficients are non-negative, the current solution

$(x_1, x_2, x_3, x_4, x_5, x_6) = (0, 0, 4, 4, 0, 0)$ is optimal.

$$Z_{opt} = \mathbf{c}_B \mathbf{x}_B = [4 \quad 3] \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 28.$$

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Solutions to Questions 4-10

Sol 4) Formulation of LP problem after introducing slack variables:

$$\text{Max. } Z = 3x_1 + 4x_2 + x_3 + 0S_1 + 0S_2 + 0S_3$$

$$\text{Subject to: } x_1 + 2x_2 + 3x_3 + S_1 = 90$$

$$2x_1 + x_2 + x_3 + S_2 = 60$$

$$3x_1 + x_2 + 2x_3 + S_3 = 80$$

$$x_1, x_2, x_3, S_1, S_2 \text{ and } S_3 \geq 0$$

Iteration 1

C _i			3	4	1	0	0	0	
	Variables	RHS	x ₁	x ₂	x ₃	S ₁	S ₂	S ₃	Ratio
0	S ₁	90	1	2	3	1	0	0	45
0	S ₂	60	2	1	1	0	1	0	60
0	S ₃	80	3	1	2	0	0	1	80
	Z _j		0	0	0	0	0	0	
	C _i -Z _j		2	4	1	0	0	0	

Iteration 2

C _i			3	4	1	0	0	0	
	Variables	RHS	x ₁	x ₂	x ₃	S ₁	S ₂	S ₃	Ratio
4	x ₂	45	1/2	1	3/2	1/2	0	0	90
0	S ₂	15	3/2	0	-1/2	-1/2	1	0	10
0	S ₃	35	5/2	0	1/2	-1/2	0	1	14
	Z _j		2	4	6	2	0	0	
	C _i -Z _j		1	0	-5	-2	0	0	

Iteration 3

C _i			3	4	1	0	0	0	
	Variables	RHS	x ₁	x ₂	x ₃	S ₁	S ₂	S ₃	
4	x ₂	40	0	1	5/3	2/3	-1/3	0	
3	x ₁	10	1	0	-1/3	-1/3	2/3	0	
0	S ₃	10	0	0	4/3	1/3	-5/3	1	
	Z _j		3	4	17/3	5/3	2/3	0	
	C _i -Z _j		0	0	-17/3	-5/3	-2/3	0	

This is a final table with optimum solution (Z=190) at x₁= 10, x₂ = 40 and x₃=0.

Sol 5). Formulation of LP problem after introducing slack variables:

$$\text{Minimize } Z = 3x_1 + 2.5x_2 - 0S_1 - 0S_2 + MA_1 + MA_2$$

$$\text{Subject to: } 2x_1 + 4x_2 - S_1 + A_1 = 40$$

$$5x_1 + 2x_2 - S_2 + A_2 = 50$$

Iteration 1

C _i			3	2.5	0	0	M	M	
	Variables	RHS	x ₁	x ₂	S ₁	S ₂	A ₁	A ₂	Ratio
M	A ₁	40	2	4	-1	0	1	0	20
M	A ₂	50	5	2	0	-1	0	1	10
	Z _j		7M	6M	-M	-M	M	M	
	C _i -Z _j		3-7M	2.5-6M	M	M	0	0	

Iteration 2

C _i			3	2.5	0	0	M	M	
	Variables	RHS	x ₁	x ₂	S ₁	S ₂	A ₁	A ₂	Ratio
M	A ₁	20	0	16/5	-1	2/5	1	-2/5	6.25
3	x ₁	10	1	2/5	0	-1/5	0	1/5	25
	Z _j		3	(16M+6)/5	-M	(2M-3)/5	M	(3-2M)/5	

	$C_i - Z_j$		0	$2.5 - [(16M+6)/5]$	M	$(3-2M)/5$	0	$(7M-3)/5$	
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Iteration 3

C_i			3	2.5	0	0	M	M
	Variables	RHS	x_1	x_2	S_1	S_2	A_1	A_2
2.5	x_2	6.25	0	1	-5/16	1/8	5/16	-1/8
3	x_1	7.5	1	0	1/8	-1/4	-1/8	1/4
	Z_j		3	2.5	-13/32	-7/16	13/32	7/16
	$C_i - Z_j$		0	0	13/32	7/16	$M - (13/32)$	$M - (7/16)$

Optimal solution: Since all $C_i - Z_j$ are ≥ 0 , the table provides the optimal solution, i.e. $x_1 = 7.5$, $x_2 = 6.25$. The optimal value of Z is 38.125

Sol 6). Formulation of LP problem after introducing slack variables:

$$\text{Max. } Z = 30x_1 + 20x_2 + 0S_1 + 0S_2 - MA_1 - MA_2$$

$$\text{Subject to: } x_1 + x_2 + S_1 = 8$$

$$6x_1 + 4x_2 - S_2 + A_1 = 12$$

$$5x_1 + 8x_2 + A_2 = 20$$

Iteration 1

C_i			30	20	0	0	-M	-M	
	Variables	RHS	x_1	x_2	S_1	S_2	A_1	A_2	Ratio
0	S_1	8	1	1	1	0	0	0	8
-M	A_1	12	6	4	0	-1	1	0	3
-M	A_2	20	5	8	0	0	0	1	5/2
	Z_j		-11M	-12M	0	M	-M	-M	
	$C_i - Z_j$		$30 + 11M$	$20 + 12M$	0	0	0	0	

Iteration 2

C_i			30	20	0	0	-M	-M	
	Variables	RHS	x_1	x_2	S_1	S_2	A_1	A_2	Ratio
0	S_1	11/2	3/8	0	1	0	0	-1/8	44/3
-M	A_1	2	7/2	0	0	-1	-1	-1/2	4/7
20	x_2	5/2	5/8	1	0	0	0	1/8	4
	Z_j		$(25-7M)/2$	20	0	M	M	$(2M+5)/2$	
	$C_i - Z_j$		$(35+7M)/2$	0	0	-M	0	-5/2	

Iteration 3

C_i			30	20	0	0	-M	-M	
	Variables	RHS	x_1	x_2	S_1	S_2	A_1	A_2	Ratio

0	S_1	37/7	0	0	1	2/28	3/28	-1/14	148/3
30	x_1	4/7	1	0	0	-2/7	-2/7	-1/7	-2
20	x_2	15/7	0	1	0	5/28	5/28	3/14	12
	Z_j		30	20	0	-20/28	-20/28	0	
	$C_i - Z_j$		0	0	0	20/28	(20-28M)/28	-M	

Iteration 4

C_i			30	20	0	0	-M	-M
	Variables	RHS	x_1	x_2	S_1	S_2	A_1	A_2
0	S_1	4	0	0	2/5	0	0	-4/35
30	x_1	4	1	8/5	0	0	0	17/35
0	S_2	12	0	28/5	0	1	1	6/5
	Z_j		30	48	0	0	0	102/7
	$C_i - Z_j$		0	-28	0	0	-M	-(7M+102)/7

Optimal solution: Since all $C_i - Z_j$ are ≤ 0 the solution is optimum at $x_1 = 4$ and $x_2 = 0$. Thus, the value of Z is 120.

Sol 7. Minimization $0.4x_1 + 0.5x_2 + MA_1 + MA_2$

Subject to $0.3x_1 + 0.1x_2 + S_1 = 1.8$

$$x_1 + x_2 + A_1 = 12$$

$$0.6x_1 + 0.4x_2 - S_2 + A_2 = 6$$

$$x_1, x_2, S_1, S_2 \geq 0$$

Where S_1 is slack variable A_1 and A_2 are artificial variable and S_2 is surplus variable.

Phase I:

Minimize Z : $A_1 + A_2$

Subject to $0.3x_1 + 0.1x_2 + S_1 = 1.8$

$$x_1 + x_2 + A_1 = 12$$

$$0.6x_1 + 0.4x_2 - S_2 + A_2 = 6$$

$$x_1, x_2, S_1, S_2 \geq 0$$

Drop the artificial variable A_1 and A_2 , we get the objective function for phase I:

Iteration 1

C_i			1.6	1.4	0	-1	0	0	
	Variables	RHS	x_1	x_2	S_1	S_2	A_1	A_2	Ratio
0	S_1	1.8	0.3	0.1	1	0	0	0	6

0	A ₁	12	1	1	0	0	1	0	12
0	A ₂	6	0.6	0.4	0	-1	0	1	10
	Z _j		0	0	0	0	0	0	
	C _i -Z _j		1.6	1.4	0	-1	0	0	

Iteration 2

C _i			1.6	1.4	0	-1	0	0	
	Variables	RHS	x ₁	x ₂	S ₁	S ₂	A ₁	A ₂	Ratio
1.6	x ₁	6	1	1/3	10/3	0	0	0	18
0	A ₁	6	0	2/3	-10/3	0	1	0	9
0	A ₂	2.4	0	1/5	-2	-1	0	1	12
	Z _j		1.6	0.53	5.3	0	0	0	
	C _i -Z _j		0	0.87	-5.3	-1	0	0	

Iteration 3

C _i			1.6	1.4	0	-1	0	0	
	Variables	RHS	x ₁	x ₂	S ₁	S ₂	A ₁	A ₂	Ratio
1.6	x ₁	3	1	0	5/3	0	-1/2	0	9/5
1.4	x ₂	9	0	1	-5	0	3/2	0	-9/5
0	A ₂	0.6	0	0	-1	-1	-0.2	1	
	Z _j		1.6	1.4	-4.33	0	1.3	0	
	C _i -Z _j		0	0	4.33	-1	0	0	

Iteration 4

C _i			1.6	1.4	0	-1	0	0	
	Variables	RHS	x ₁	x ₂	S ₁	S ₂	A ₁	A ₂	
0	S ₁	9/5	3/5	0	1	0	-3/10	0	
1.4	x ₂	18	3	1	0	0	0	0	
0	A ₂	0.6	3/5	0	0	-1	-0.5	1	
	Z _j		4.2	1.4	0	0	0	0	
	C _i -Z _j		-2.6	0	0	-1	0	0	

This is the final table and A₂ is still present in the basis which makes this L P infeasible.

Sol 8.

$$\max Z = 2x_1 + 3x_2 + x_3 - 0S_1 - 0S_2 - 0S_3 + MA_1 + MA_2$$

$$\text{Subject to } x_1 + x_2 + x_3 + S_1 = 40$$

$$2x_1 + x_2 - x_3 - S_2 + A_1 = 10$$

$$-x_2 + x_3 - S_3 + A_2 = 10$$

Phase I:

Iteration 1

C _i			2	0	0	0	-1	-1	0	0	
	Variables	RHS	x ₁	x ₂	x ₃	S ₁	S ₂	S ₃	A ₁	A ₂	Ratio
0	S ₁	40	1	1	1	1	0	0	0	0	40
0	A ₁	10	2	1	-1	0	-1	0	1	0	5
0	A ₂	10	0	-1	1	0	0	-1	0	1	∞
	Z _j		0	0	0	0	0	0	0	0	
	C _i -Z _j		2	0	0	0	-1	-1	0	0	

Iteration 2

C _i			2	0	0	0	-1	-1	0	0	
	Variables	RHS	x ₁	x ₂	x ₃	S ₁	S ₂	S ₃	A ₁	A ₂	Ratio
0	S ₁	35	0	1/2	3/2	1	1/2	0	-1/2	0	23.3
2	x ₁	5	1	1/2	-1/2	0	-1/2	0	1/2	0	-10
0	A ₂	10	0	-1	1	0	0	-1	0	1	10
	Z _j		2	1	-1	0	1	0	1	0	
	C _i -Z _j		0	-1	1	0	-2	-1	-1	0	

Iteration 3

C _i			2	0	0	0	-1	-1	0	0	
	Variables	RHS	x ₁	x ₂	x ₃	S ₁	S ₂	S ₃	A ₁	A ₂	Ratio
0	S ₁	20	0	2	0	1	1/2	3/2	-1/2	-3/2	
2	x ₁	10	1	0	0	0	-1/2	-1/2	1/2	1/2	
0	x ₃	10	0	-1	1	0	0	-1	0	1	
	Z _j		2	0	0	0	-1	-1	1	1	
	C _i -Z _j		0	0	0	0	0	0	-1	-1	

The optimal value of the Phase I problem is $w = 0$. So the original problem is feasible, and a basic feasible solution is $x_1 = 10$; $x_3 = 10$; $S_1 = 20$; $x_2 = S_2 = S_3 = 0$. Now we can start Phase

II. Again the objective value z should be represented by the non-basic variables:

$$z = 2x_1 + 3x_2 + x_3 = 30 + 4x_2 + S_2 + 2S_3$$

The initial tableau is (the last Phase I tableau with A₁ and A₂ taken away):

C _i			0	4	0	0	1	2	
	Variables	RHS	x ₁	x ₂	x ₃	S ₁	S ₂	S ₃	Ratio
0	S ₁	20	0	2	0	1	1/2	3/2	10

0	x_1	10	1	0	0	0	-1/2	-1/2	∞
0	x_3	10	0	-1	1	0	0	-1	-10
	Z_j		0	0	0	0	0	0	
	$C_i - Z_j$		0	4	0	0	1	2	

Iteration 4

C_i			0	4	0	0	1	2
	Variables	RHS	x_1	x_2	x_3	S_1	S_2	S_3
4	x_2	10	0	1	0	1/2	1/4	3/4
0	x_1	10	1	0	0	0	-1/2	-1/2
0	x_3	20	0	0	1	1/2	1/4	-1/4
	Z_j		0	4	0	2	1	3
	$C_i - Z_j$		0	0	0	-2	0	-1

Optimal solution: $x_1 = 10$; $x_2 = 10$; $x_3 = 20$; $Z = 70$

Sol 9. Formulation of LP problem after introducing slack variables

$$\text{Max. } Z = 3x_1 + x_2 + 0x_3 + 0S_1 + 0S_2 + 0S_3$$

$$\text{Subject to: } x_1 + 2x_2 + S_1 = 5$$

$$x_1 + x_2 - x_3 + S_2 = 2$$

$$7x_1 + 3x_2 - 5x_3 + S_3 = 20$$

Iteration 1

C_i			3	1	0	0	0	0	
	Variables	RHS	x_1	x_2	x_3	S_1	S_2	S_3	Ratio
0	S_1	5	1	2	0	1	0	0	5
0	S_2	2	1	1	-1	0	1	0	2
0	S_3	20	7	3	-5	0	0	1	20/7
	Z_j		0	0	0	0	0	0	
	$C_i - Z_j$		3	1	0	0	0	0	

Iteration 2

C_i			3	1	0	0	0	0	
	Variables	RHS	x_1	x_2	x_3	S_1	S_2	S_3	Ratio
0	S_1	3	0	1	1	1	-1	0	3
3	x_1	2	1	1	-1	0	1	0	-2
0	S_3	6	0	-4	2	0	-1	1	3
	Z_j		3	3	-3	0	3	0	
	$C_i - Z_j$		0	-2	3	0	-3	0	

Iteration 3

C _i			3	1	0	0	0	0
	Variables	RHS	x ₁	x ₂	x ₃	S ₁	S ₂	S ₃
0	x ₃	3	0	1	1	1	-1	0
3	x ₁	5	1	2	0	1	0	0
0	S ₃	0	0	-6	0	-2	-5	1
	Z _j		3	6	0	3	0	0
	C _i -Z _j		0	-5	0	-3	0	0

Optimal solution: $x_1 = 5$; $x_2 = 0$; $x_3 = 3$; $Z=15$

Sol 10. Minimize $Z = -x_1 + x_2 + 0S_1 + 0S_2 + 0S_3 + MA_1 + MA_2$

Subject to: $x_1 - 4x_2 - S_1 + A_1 = 5$

$x_1 - 3x_2 + S_2 = 1$

$2x_1 - 5x_2 - S_3 + A_2 = 1$

Iteration 1

C _i			-1	1	0	0	0	M	M	Ratio
	Variables	RHS	x ₁	x ₂	S ₁	S ₂	S ₃	A ₁	A ₂	
M	A ₁	5	1	-4	-1	0	0	1	0	5
0	S ₂	1	1	-3	0	1	0	0	0	1
M	A ₂	1	2	-5	0	0	-1	0	1	1/2
	Z _j		3M	-9M	-M	0	-M	M	M	
	C _i -Z _j		-1-3M	1+9M	M	0	M	0	0	

Iteration 2

C _i			-1	1	0	0	0	M	M	
	Variables	RHS	x ₁	x ₂	S ₁	S ₂	S ₃	A ₁	A ₂	Ratio
M	A ₁	9/2	0	-3/2	-1	0	1/2	1	-1/2	9
0	S ₂	1/2	0	-1/2	0	1	1/2	0	-1/2	1
-1	x ₁	1/2	1	-5/2	0	0	-1/2	0	1/2	-1
	Z _j		-1	(5-3M)/2	-M	0	(M-1)/2	M	(1-M)/2	
	C _i -Z _j		0	(3M-3)/2	M	0	-(M+1)/2	0	(3M+1)/2	

Iteration 3

C _i			-1	1	0	0	0	M	M
	Variables	RHS	x ₁	x ₂	S ₁	S ₂	S ₃	A ₁	A ₂
M	A ₁	4	0	-1	-1	-1	0	1	0
0	S ₃	1	0	-1	0	2	1	0	-1
-1	x ₁	1	1	-3	0	1	0	0	0

	Z_j		-1	$3-M$	$-M$	$-(1+M)$	0	M	0
	C_i-Z_j		0	$M-2$	M	$M+1$	0	0	M

Optimality conditions are satisfied but presence of artificial variable in the final table indicates infeasibility.